PlateMate: Automated Nutrient Tracking System

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Motivation

As global health trends shift toward mindful eating, the prevalence of **junk food** remains a pressing issue, especially among **children**. In schools, many children are not provided with adequately balanced, **nutritious meals**, leaving them vulnerable to poor dietary habits as they often choose junk food over healthier options, becoming a root cause of many health issues.

Solution

PlateMate is an intelligent mobile app that scans food plates to analyze their nutritional content. By leveraging Machine Learning and TensorFlow, PlateMate provides a user-friendly platform to track key nutrients such as calories, carbohydrates, protein, and sugars. This innovation empowers individuals, especially parents and educators, to make healthier dietary decisions for their children.

PlateMate Workflow

Diverse Dataset Collection (6 categories, angles) → Preprocessing (rescaling, rotation, flipping) → Model Training (MobileNetV2 architecture, optimized training parameters) → App Deployment (upload plate image, receive nutritional breakdown)

Data Collection

- 480 total images captured
- (6 food categories) x (80 images per category)
- Apples (red and green apples), Oranges, Bananas, Chips (6 flavors), Cookies (3 flavors), Yogurt (3 flavors)
- Variety of Angles and Orientations: objects photographed from multiple angles (front, side, top-down, tilted), ensuring the model can detect items even in varied orientations
- Data Augmentation Techniques: rescaling, rotation, shift, zoom, flipping
- Dataset Split: Training (70%), Validation (15%), Testing (15%)



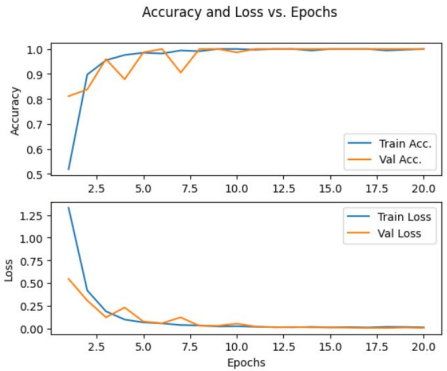
Model Architecture

- **Layer 1:** MobileNetV2 (Transfer Learning), pre-trained on ImageNet dataset for feature extraction
- **Layer 2:** Global Average Pooling Layer, reduces spatial dimensions of feature maps
- **Layer 3:** Dropout Rate 20%, neurons randomly dropped to prevent overfitting
- **Layer 4:** Fully Connected Dense, ReLU activation for non-linearity to learn complex patterns
- **Layer 5:** Output Dense, 6 neurons corresponding to the 6 food categories being classified (Softmax activation)
- **Quantization**: Reduced model size from 6.3 MB to 0.79 MB using TensorFlow Lite.

Layer (type)	Output Shape	Param #
mobilenetv2_0.35_224 (Functional)	(None, 7, 7, 1280)	410,208
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 1280)	0
dropout_1 (Dropout)	(None, 1280)	0
dense_2 (Dense)	(None, 128)	163,968
dense_3 (Dense)	(None, 6)	774

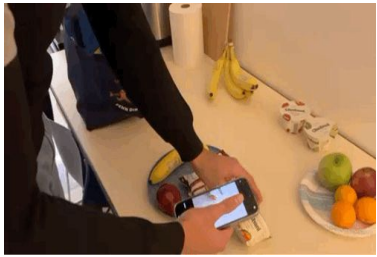
Training & Results

- 20 Epochs
- Adam Optimizer
- Categorical Cross Entropy Loss



Final Training Accuracy, Loss: 100%, 0.0116
Final Validation Accuracy, Loss: 100%, 0.0054
Final Test Accuracy, Loss: 100%, 0.00249

Hardware Deployment Demo



Real-time food recognition using PlateMate.
Try yourself! https://moshah122.github.io/PlateMate/

Future Work

- Integrate with wearable devices.
- Expand the dataset to include mixed and processed food items.
- Incorporate additional metrics, such as micronutrient tracking.